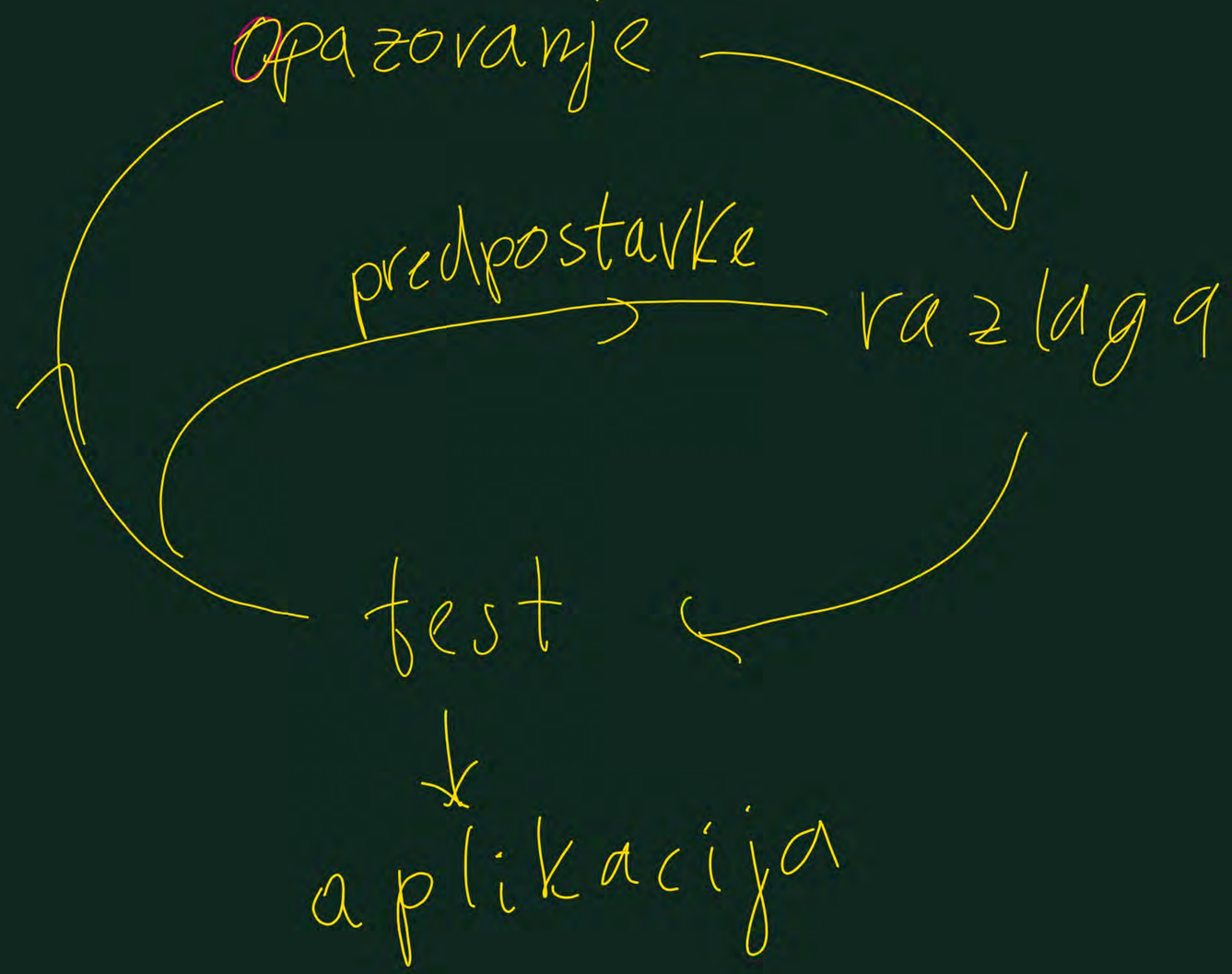

opazovanje
predpostavke
razlaga
test
aplikacija

Kinematika

opazovalec → opazovalni sistem

- izhodišče
- ura → čas $[t] = s$ sekunda

predpona μs $10^{-6} = 1\mu$

$0{,}000001\ s = 1\ \mu s$

- koordinatna os razdalja, koordinata x

$[x] = m$ meter

Premo gibanje



koordinatna os

lega (koordinata x)

$x(t) = f(t)$

premik

$\vec{d}$

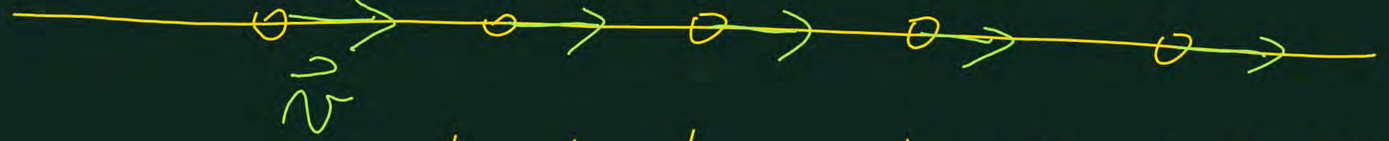
$x(t_1) = x_1$

razdalja

$|\vec{d}| = d$

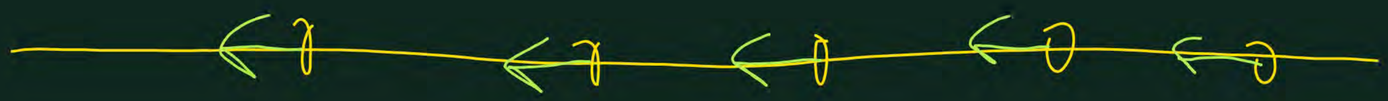
pot

Diagram ~~lege~~ gibanja (reprezentacija)



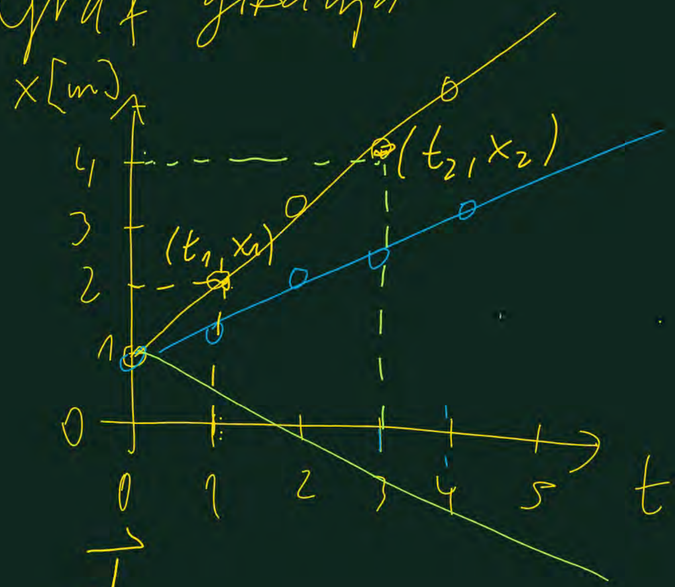
enakomerno

$t-\Delta t$ t $t+\Delta t$ $t+2\Delta t$ (premo) gibanje



x [m]	t [s]	x
1	0	1
2	1	1,5
3	2	2
4	3	2,5
5	4	3

Graf gibanja



$$k = \frac{x_2 - x_1}{t_2 - t_1} =$$

$$v_x = \frac{\Delta x}{\Delta t}$$

$\Delta x < 0$

$v_x < 0$

$$\vec{v} = \frac{\vec{d}}{\Delta t}$$

Delta $x = x_{naraven} - x_{začetka}$

$$[v] = \frac{[\Delta x]}{[\Delta t]} = \frac{m}{s}$$

$$\Delta x = v_x \Delta t$$

$$x - x_0 = v_x t$$

$t_1 \to 0 \quad x_1 \to x_0$

$t_2 \to t \quad x_2 \to x$

$$x = x_0 + v_x t$$

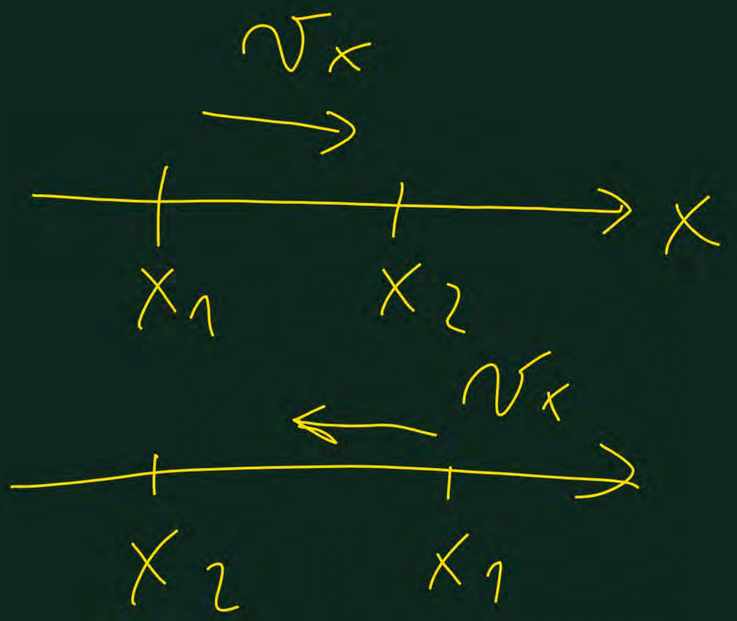
Predznak hitrosti

$$v_x = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

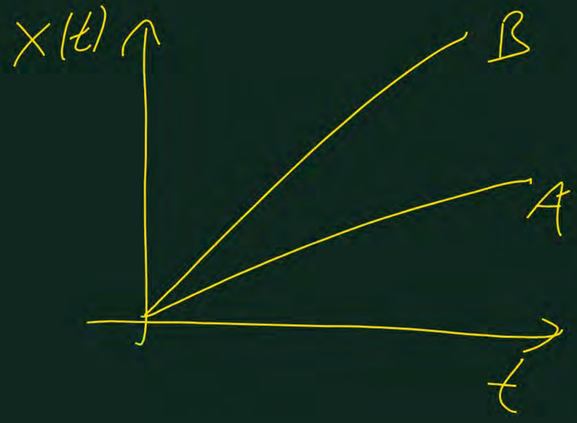
$\Delta t > 0$

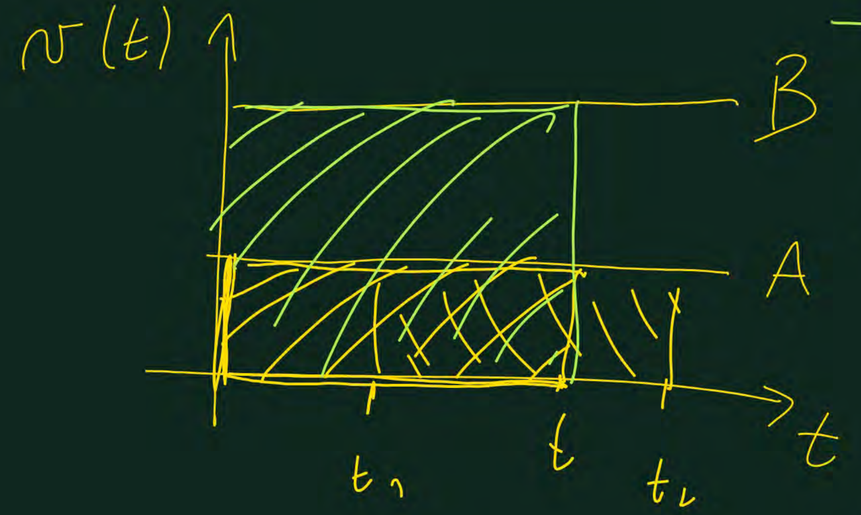
$x_2 > x_1 \; ; \; x_2 - x_1 > 0 \; ; \; v_x > 0$

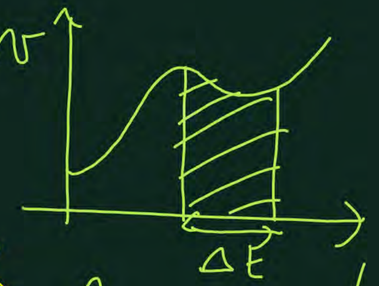
$x_2 < x_1 \; ; \; x_2 - x_1 < 0 \; ; \; v_x < 0$

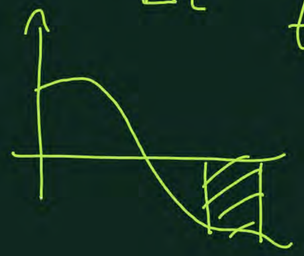


Graf $v(t)$







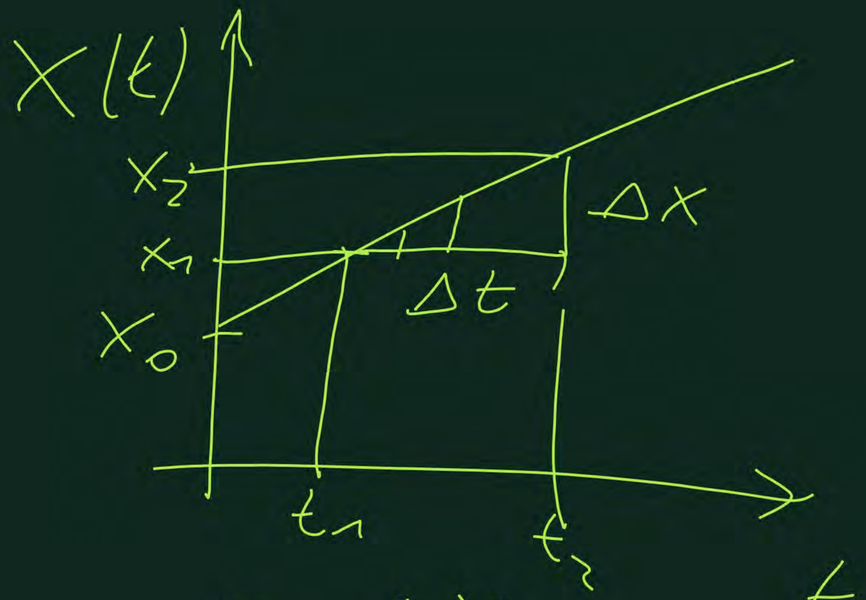


$$x = x_0 + v_x t \rightarrow \underline{x - x_0 = v_x t}$$

$$[v][t] = \frac{m}{p} \cdot p = m$$

$$x_2 - x_1 = v_x (t_2 - t_1)$$

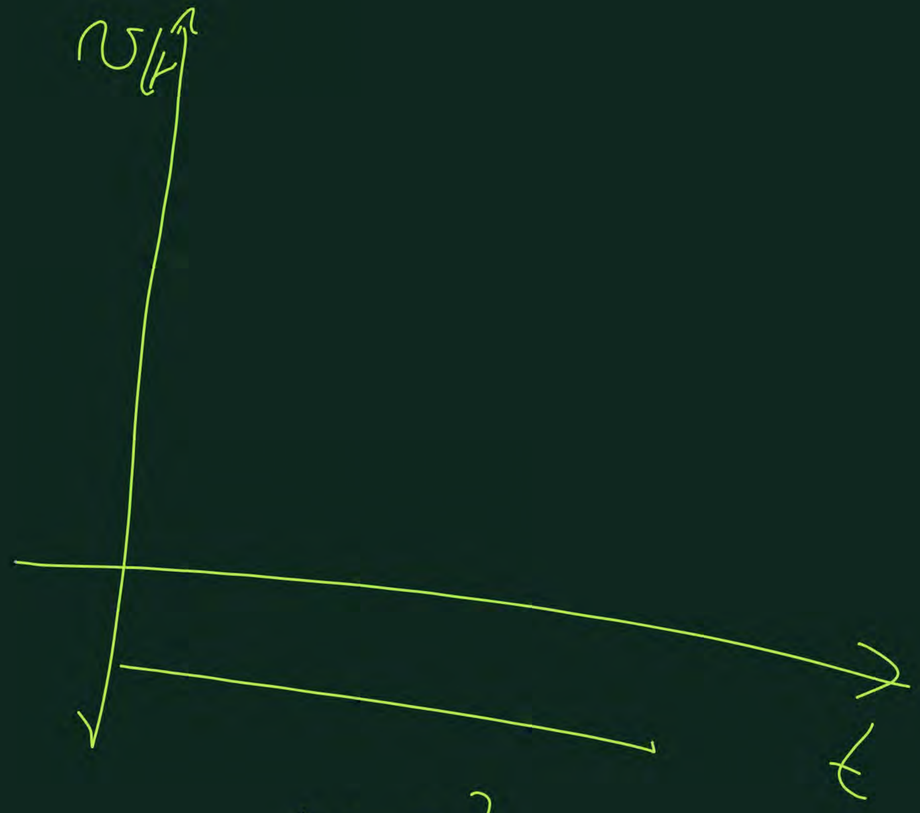
$$x - x_0 = v_x t$$



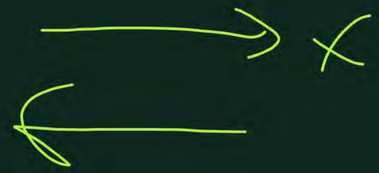
$$v_x = \frac{\Delta x}{\Delta t}$$

$\Delta t \to 0$

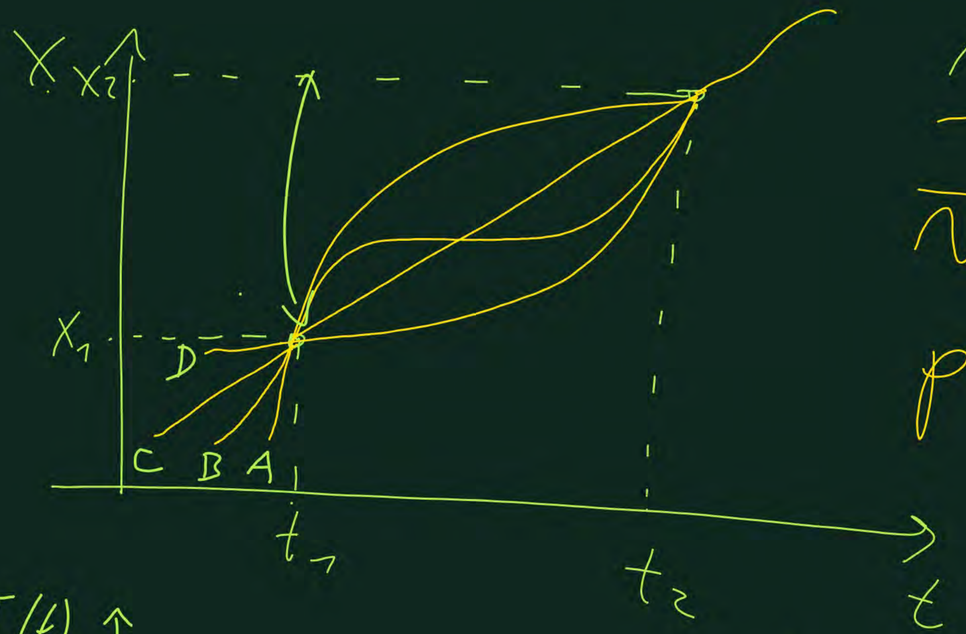




$x_0 = ?$



Povprečna hitrost

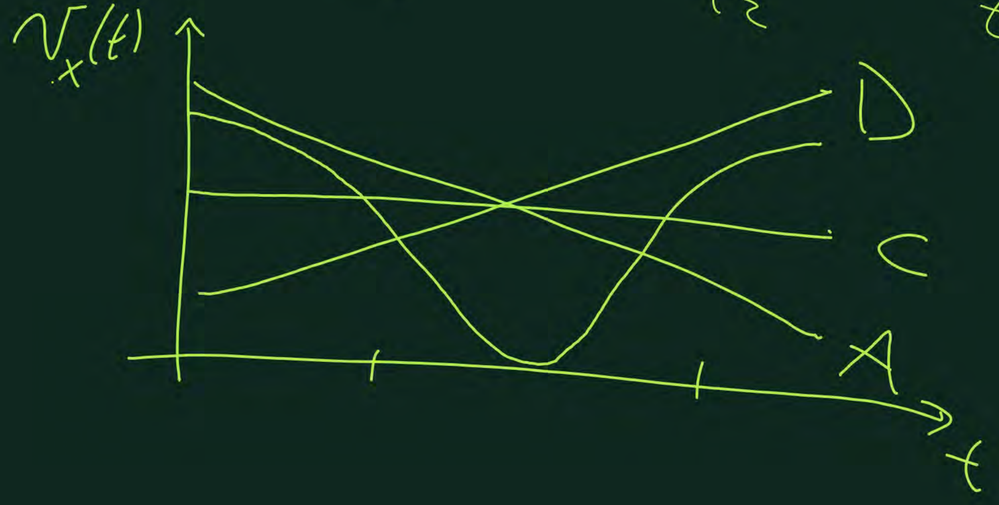


$$\overline{v_x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

$$\overline{v_x} = \frac{\Delta x}{\Delta t}$$

povprečna hitrost

velikost hitrosti $v_v = \frac{D}{t}$



diferenca

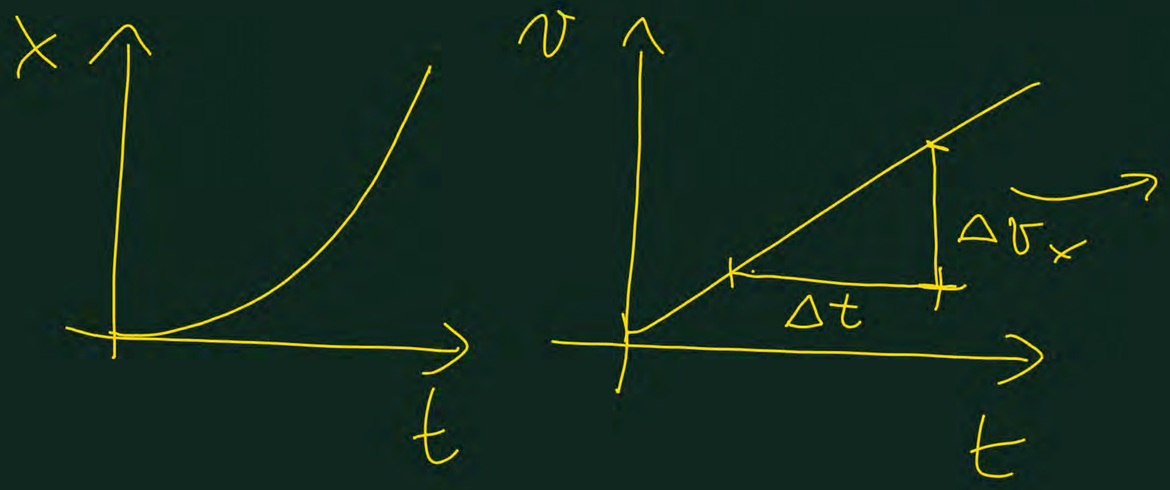
$\Delta t \to 0$

diferencial

dt

$v_x = \frac{dx}{dt}$ → trenutna hitrost

Prosti pad



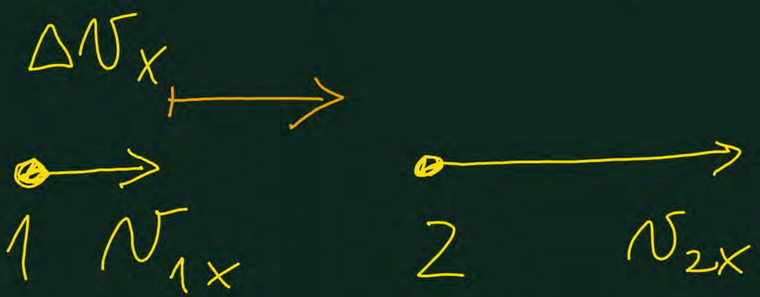
strmina pospešek

$$a_x = \frac{\Delta v_x}{\Delta t}$$

$$[a] = \frac{m}{s^2} = \frac{m/s}{s}$$

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$$

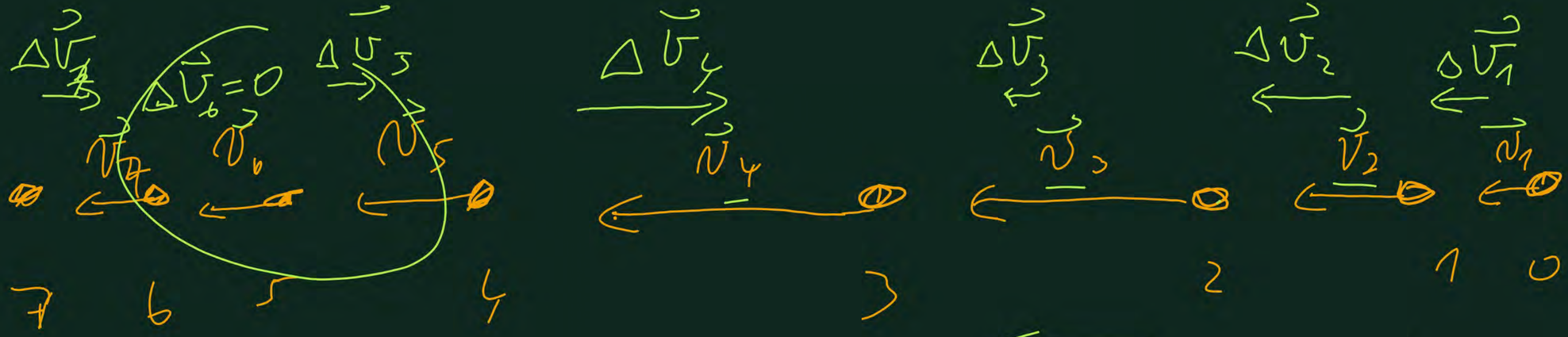
Diagram gibanya

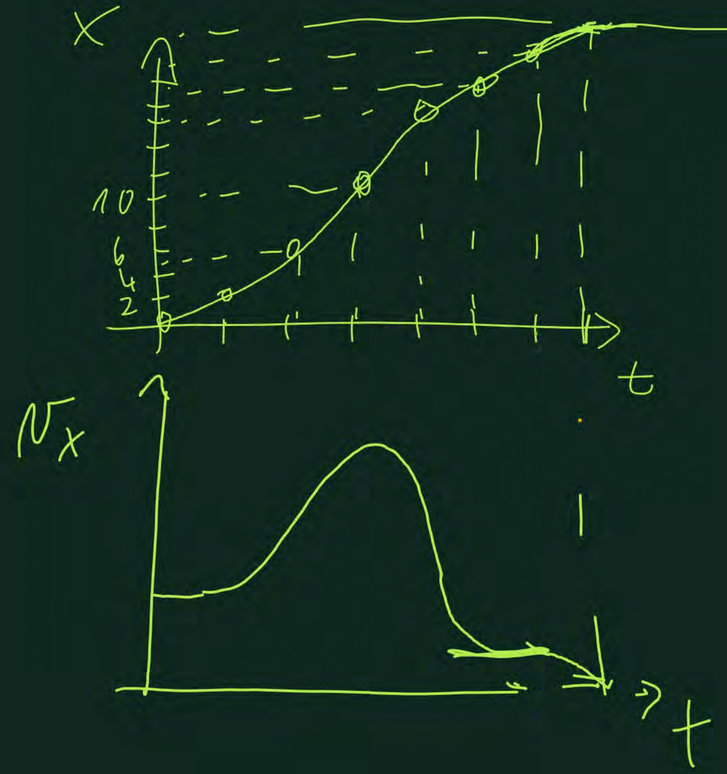


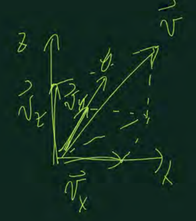


$$\Delta v_x = v_{2x} - v_{1x}$$

$$v_{1x} + \Delta v_x = v_{2x}$$







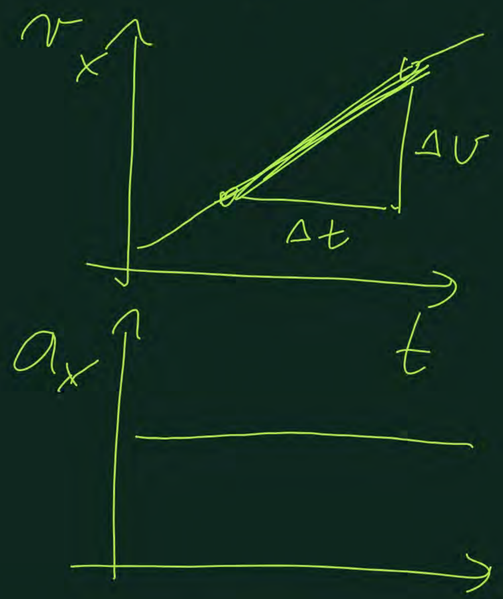
$\vec{v} = (v_x, v_y, v_z)$

$\vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z =$

$= v_x \hat{e}_x + v_y \hat{e}_y + v_z \hat{e}_z$

Kvantitativna analiza pospeška

$$a_x = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1}$$



$$v_{2x} - v_{1x} = a_x(t_2 - t_1)$$

$v_{1x} = v_{0x}$ (začetna hitrost)

$t_1 = 0$

$v_{2x} \to v_x$

$t_2 \to t$

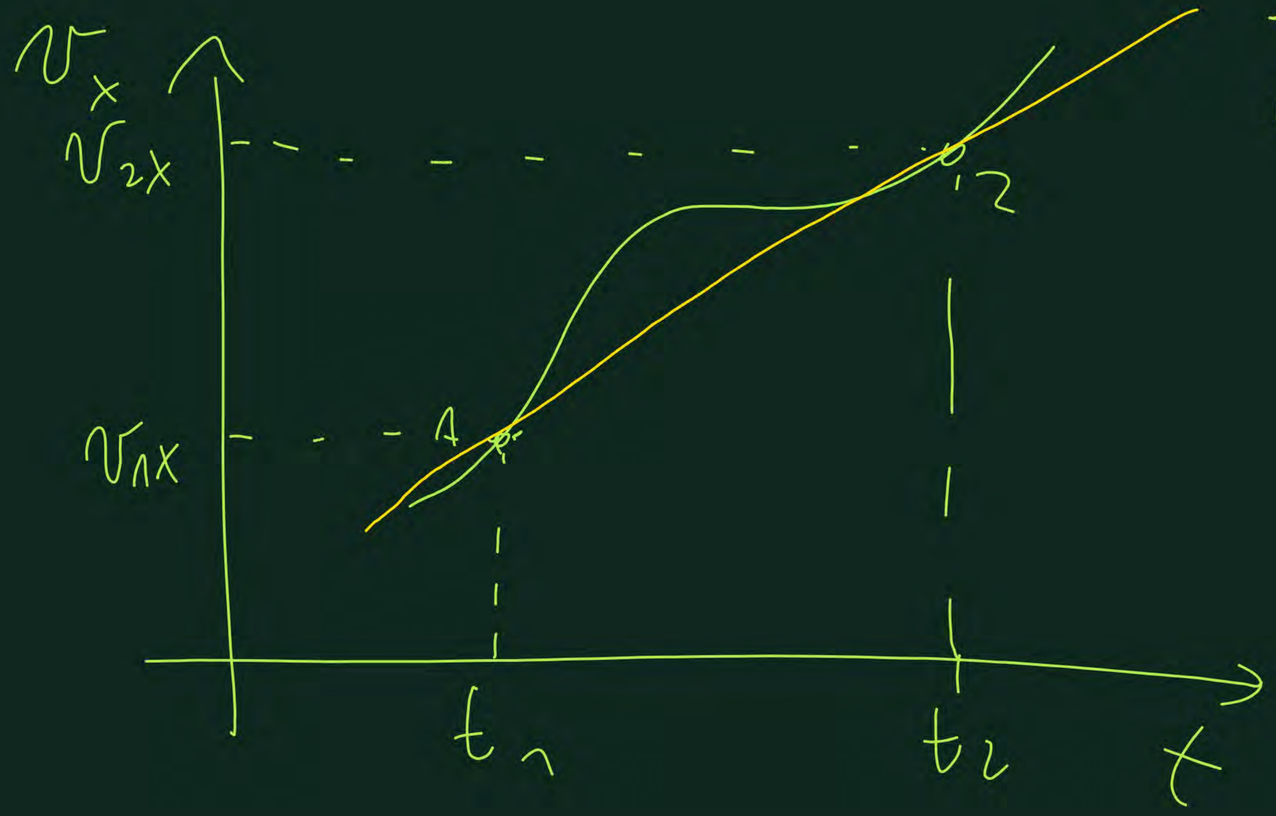
$$\boxed{v_x = v_{0x} + a_x t}$$

enakomerno pospešeno gibanje

a_x = konst.

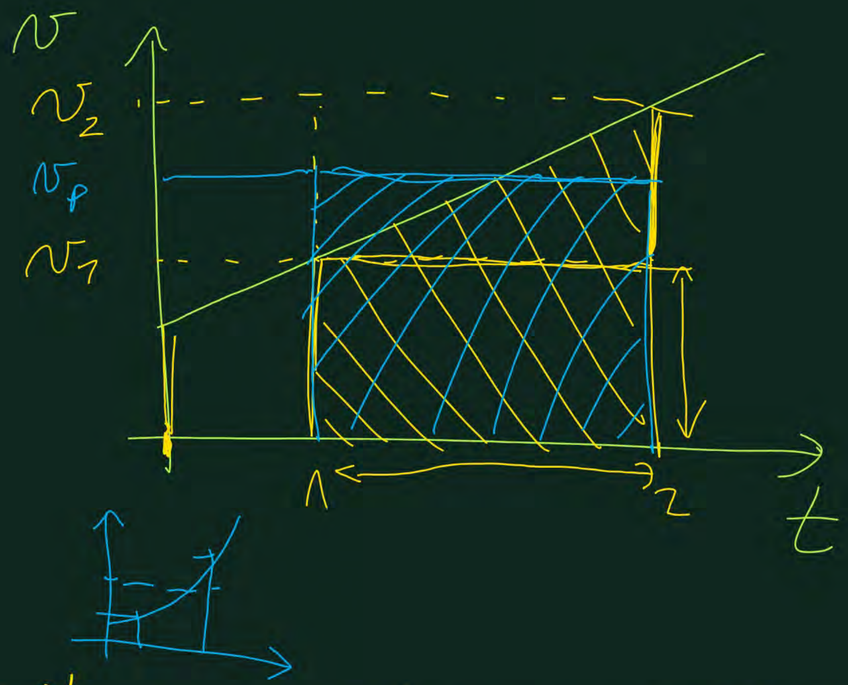


Povprečni pospešek



$$\overline{a_x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1}$$

Povprečna hitrost (pri enakomerno posp.g.)



$$x_2 - x_1 = (t_2 - t_1) v_1 + \frac{1}{2}(t_2 - t_1)(v_2 - v_1)$$

$$= (t_2 - t_1)\frac{1}{2}(v_1 + v_2)$$

$$x_2 - x_1 = v_p (t_2 - t_1)$$

$$v_p = \frac{v_1 + v_2}{2}$$

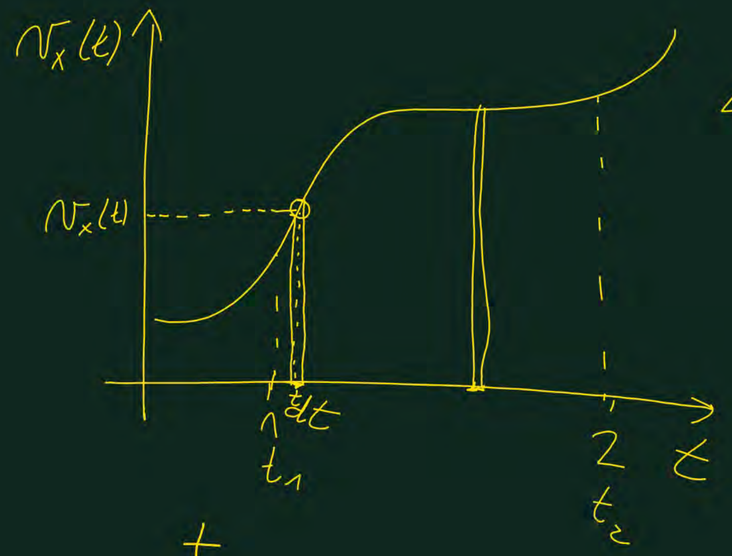
Premik pri enakomerno pos.p.g.

$t_1 \to 0$
$t_2 \to t$
$x_1 \to x_0$
$x_2 \to x$

$$v_x = v_{0x} + a_x t$$

$$x - x_0 = t\,\frac{1}{2}(v_{0x} + v_{0x} + a_x t) =$$

$$\boxed{x - x_0 = v_{0x} t + \frac{1}{2} a_x t^2}$$



$$\Delta x = \int_{t_1}^{t_2} v_x(t)\,dt$$

$$\sum \longrightarrow \int$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$v_x = v_{x0} + at$$

$$\Delta x = \int_0^t (v_{x0} + a_x t')\,dt' = \int_{t_0}^t v_{x0}\,dt' + \int_0^t a_x t'\,dt' =$$

$$= v_{x0}\frac{t'^1}{1}\Big|_0^t + a_x\frac{t'^2}{2}\Big|_{t_0} =$$

$$(x^{n+1})' = (n+1)x^n$$

$$\int (f+g)\,dx =$$

$$= \int f\,dx + \int g\,dx$$

$$\int c\,f(x)\,dx = c\int f(x)\,dx$$

$$x - x_0 = v_{x0}t - v_{x0}0 + \frac{1}{2}a_x t^2 - \frac{1}{2}a_x 0^2 = v_{x0}t + \frac{1}{2}a_x t^2$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad \rightarrow x(t)$$

$$v_x = v_{0x} + a_x t \quad \rightarrow v(t)$$

; $v(x) = ?$

$$t = \frac{v_x - v_{0x}}{a_x}$$

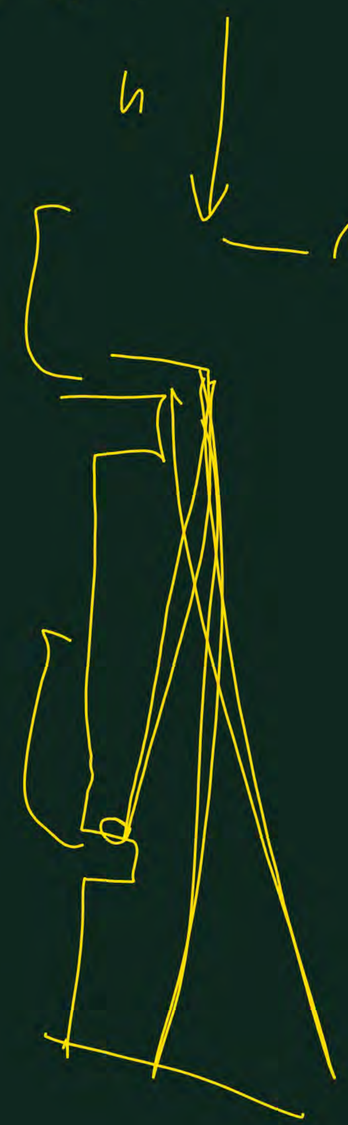
$$x = x_0 + v_{0x}\frac{v_x - v_{0x}}{a_x} + \frac{1}{2}a_x\frac{(v_x - v_{0x})^2}{a_x^2} \quad \Big| \cdot 2 \cdot a_x$$

$$2(x - x_0)a_x = 2v_{0x}v_x - 2v_{0x}^2 + v_x^2 + v_{0x}^2 - 2v_x v_{0x}$$

$$\boxed{2a_x(x - x_0) = v_x^2 - v_{0x}^2}$$

$$\boxed{v_x = \pm\sqrt{v_{0x}^2 + 2a_x(x - x_0)}}$$

Poskus:



$v_x = 11{,}2$ m/s — 6,41 m

$v_x = 10{,}4$ m/s — h = 6,357 m

$v_x = 10{,}44$ m/s — = 6357 mm

$v_x = (11{,}2)$ m/s



$(10{,}4 - 11{,}2)$ m/s

$(10{,}8 \pm 0{,}4)$ m/s